Stage 8 Science — Practice Test 3

Paper 1: 45 minutes · calculator allowed · Paper 2: 45 minutes · Total 100 marks

Answer all questions. Write your answers in the spaces provided. Show your working where there is space for it. The number of marks for each question is shown in brackets [ ].

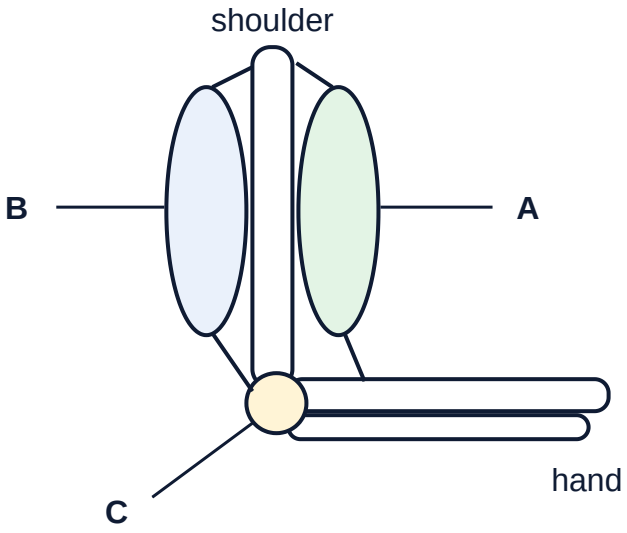
Name _____ Class _____ Date _____ Mark: ____ / 100

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 45 minutes — 50 marks

1 Amir trains at a gym and wears a fitness tracker on his wrist. **[1]**

The diagram shows the bones and two muscles of his arm.



(a) Amir bends his arm at the elbow. What type of joint is the elbow?

- ☐ A) ball-and-socket joint
- ☐ B) fixed joint
- ☐ C) hinge joint
- ☐ D) a ligament

(b) Amir can swing his arm round in a full circle at the shoulder. Name the type of joint at the shoulder. **[1]**

(c) Amir lifts a dumbbell by bending his arm at the elbow. Which letter, **A** or **B**, shows the muscle that contracts? **[1]**

(d) Explain why the muscles at a hinge joint must work as an **antagonistic pair**. **[2]**

(e) The fitness tracker shines light into Amir's skin to measure how much oxygen his blood is carrying. Which part of the blood transports oxygen? **[1]**

- ☐ A) plasma
- ☐ B) red blood cells
- ☐ C) white blood cells
- ☐ D) carbon dioxide

2 Sara investigates how the speed of a treadmill affects her breathing rate. **[1]**

She runs for 3 minutes at each speed, then her fitness tracker counts her breaths in one minute. She repeats each speed three times on different days.

Speed in km/h	Test 1 (breaths per minute)	Test 2	Test 3	Mean
4	18	17	19	18
6	24	25	23	24
8	30	46	32	?
10	36	37	38	37

(a) Name the **independent** variable in Sara's investigation.

(b) Give **one** variable Sara should keep the same to make her results valid. **[1]**

(c) One result in the table is anomalous. Write down this result. **[1]**

Working

Answer ____ breaths per minute

(d) Calculate the mean breathing rate at 8 km/h. Do **not** include the anomalous result. Show your working. **[2]**

Working

Answer _____ breaths per minute

(e) The results show that Sara breathes faster when she runs faster. **[2]**
Use ideas about **respiration** to explain why.

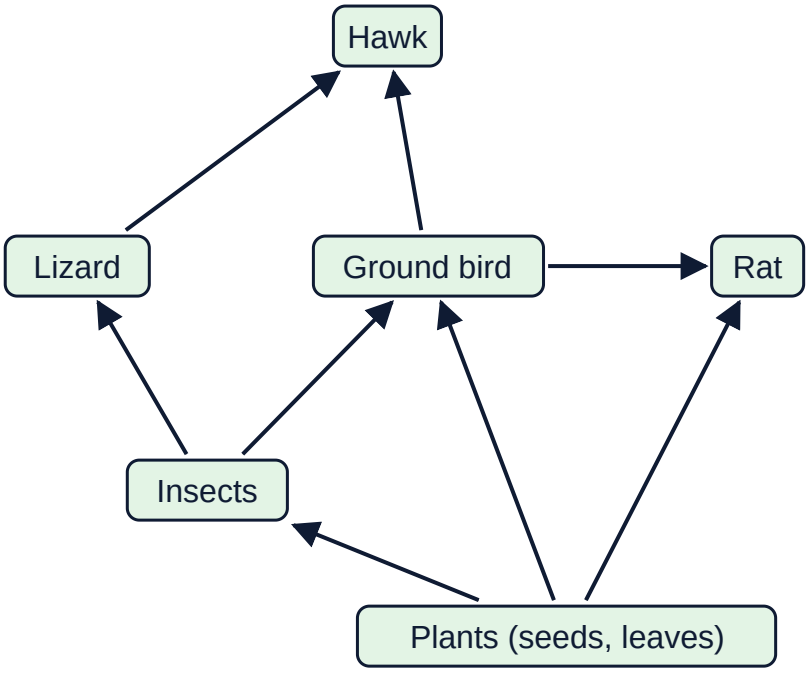
(f) Why did Sara repeat each speed three times? **[1]**

- ☐ A) to make the treadmill go faster
- ☐ B) to change the independent variable
- ☐ C) to have more control variables
- ☐ D) to spot anomalous results and make the results more reliable

3 A small island has a forest, a sandy beach and rock pools. **[1]**

Rats arrived on the island by accident on a boat. There were no rats on the island before. Rats eat seeds and also eat the eggs of ground birds.

The diagram shows part of the island's food web after the rats arrived.



(a) The island is an ecosystem. Name **one** habitat on the island.

(b) Rats are an **invasive species** on this island. What does this mean? **[1]**

- ☐ A) a species that has always lived on the island
- ☐ B) a species that only eats plants
- ☐ C) a new species brought to an area that spreads and harms the organisms already there
- ☐ D) a species that has died out

(c) After the rats arrived, the number of ground birds went down. **[2]**

Use the food web to give **two** reasons.

(d) Scientists then found that the number of insects **increased**. Use the food web to explain why. **[1]**

4 A factory makes fertilisers for farms. **[1]**

One fertiliser, ammonium sulfate, is made by reacting ammonia with sulfuric acid. The reaction makes only one product.

(a) Complete the word equation.

ammonia + sulfuric acid → _____

(b) The solid that leaves the factory also contains some water and a little unreacted acid. Which statement describes this solid? **[1]**

- ☐ A) It is impure, because it is a mixture of more than one substance.
- ☐ B) It is pure, because it was made in a chemical reaction.
- ☐ C) It is pure, because it is a solid.
- ☐ D) It is an element.

(c) A quality-control worker tests a 200 g sample of the dry fertiliser. It contains 190 g of ammonium sulfate. **[2]**

Calculate the percentage purity of the sample. Show your working.

Working

Answer ☐ %

(d) The factory makes a second fertiliser by reacting potassium hydroxide (an alkali) with nitric acid. **[1]**

Complete the word equation.

potassium hydroxide + nitric acid → potassium nitrate + _____

5 On a space mission, tanks of helium gas are used to push fuel into the rocket engines. **[1]**

A helium atom has 2 protons, 2 neutrons and 2 electrons.

(a) What is the charge on a neutron?

- ☐ A) positive
- ☐ B) negative
- ☐ C) no charge
- ☐ D) it is positive in some atoms and negative in others

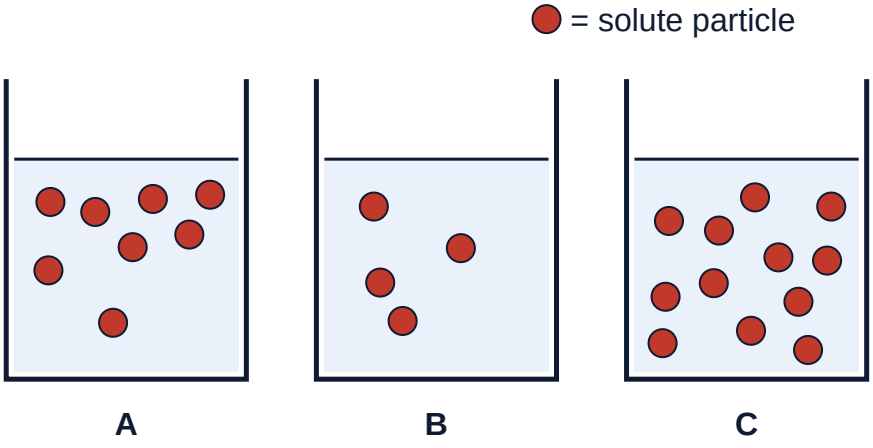
(b) What holds the electrons in the helium atom? **[1]**

- ☐ A) gravity between the electrons
- ☐ B) the electrostatic attraction between the positive nucleus and the negative electrons
- ☐ C) the magnetism of the neutrons
- ☐ D) friction between the particles

(c) Describe the Rutherford model of the atom. **[2]**

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6 The factory also sells a liquid fertiliser (a solution of a salt in water). **[1]**
The diagram shows the solute particles in three samples, A, B and C. Each beaker has the **same volume** of water.



(a) Which sample, **A**, **B** or **C**, is the most concentrated?

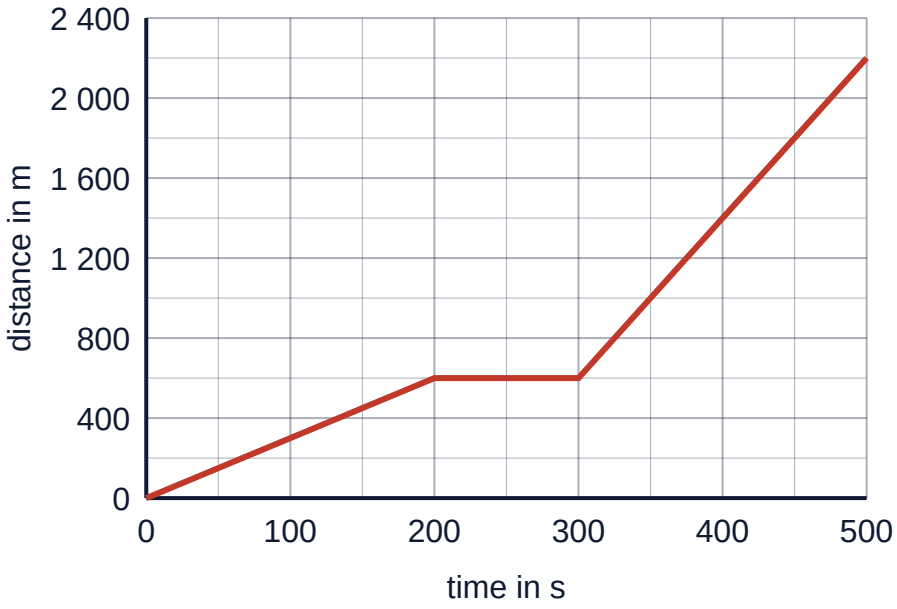
(b) A farmer adds more water to sample C. What happens to the concentration of the solution? **[1]**

- ☐ A) it decreases
- ☐ B) it increases
- ☐ C) it stays the same
- ☐ D) it becomes zero

(c) Use ideas about particles to explain your answer to (b). **[1]**

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7 Yusuf cycles up a hill, rests at the top, and then cycles down the other side. **[2]**
The distance–time graph shows his journey.



(a) Calculate Yusuf's speed going **up** the hill (the first 200 s). Show your working.

Working

Answer ____ m/s

(b) Describe Yusuf's motion between 200 s and 300 s. **[1]**

(c) During which part of the journey did Yusuf travel fastest? Explain how you can tell from the graph. **[1]**

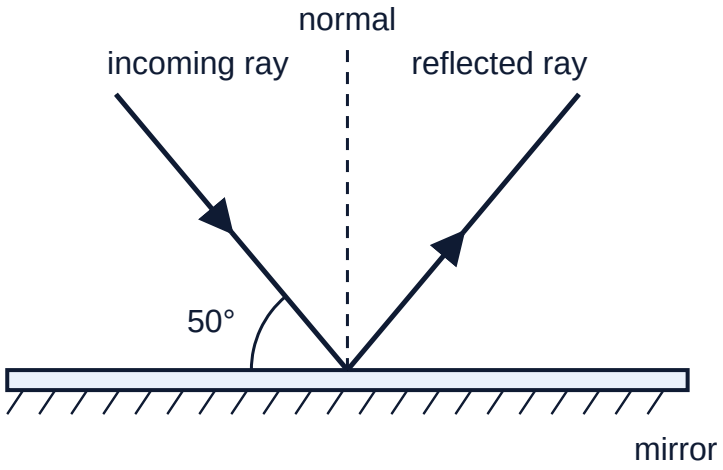
(d) Calculate Yusuf's **average** speed for the whole journey of 500 s. Show your working. **[2]**

Working

Answer ____ m/s

8 A theatre uses a flat mirror to send a spotlight beam onto the stage. **[1]**

The diagram shows a ray of light hitting the mirror. The angle between the incoming ray and the mirror is 50°.



(a) What is the **angle of incidence**?

Working

Answer °

(b) What is the **angle of reflection**? **[1]**

Working

Answer °

(c) An actor wears a red costume. White light shines on it. **[2]**

Explain why the costume looks red.

(d) The stage lights change so that only **blue** light shines on the red costume. What colour does the costume look now? **[1]**

- ☐ A) red
- ☐ B) blue
- ☐ C) magenta
- ☐ D) black

(e) Two spotlights, one red and one green, overlap on a white screen. What colour is seen where they overlap? **[1]**

_____

9 At the fertiliser factory, salt crystals move along a conveyor belt. An electromagnet above the belt pulls out any small pieces of iron that have fallen into the salt. **[2]**

(a) Describe how to make a simple electromagnet.

_____

_____

_____

_____

(b) Why does the factory use an electromagnet and not a permanent magnet? **[1]**

- ☐ A) it can be switched off, so the iron pieces drop off into a bin
- ☐ B) it also attracts the salt
- ☐ C) it never needs electricity
- ☐ D) it attracts copper and aluminium

10 A space agency is planning a mission to take pictures of a nearby galaxy. **[1]**

(a) Which list describes what a galaxy contains?

- ☐ A) only planets
- ☐ B) one star and the planets that orbit it
- ☐ C) only dust and gas
- ☐ D) billions of stars, planetary systems, and dust and gas

(b) Before launch, engineers test the spacecraft's compass on Earth. The compass needle lines up with the Earth's magnetic field. **[1]**
Why does the Earth have a magnetic field?

- ☐ A) the Moon pulls on it
- ☐ B) the Earth's core acts as a magnet
- ☐ C) sunlight makes the air magnetic
- ☐ D) the oceans are magnetic

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11 Students in a climate data project collect information about their town. **[1]**

(a) Which statement describes the **climate** of the town, not the weather?

- ☐ A) It rained heavily this morning.
- ☐ B) Yesterday's highest temperature was 31 °C.
- ☐ C) Summers are usually hot and dry, based on 30 years of records.
- ☐ D) There is a strong wind today.

(b) The school's weather station gets its electricity from a solar panel. **[1]**
Is solar power a renewable or a non-renewable resource? Give a reason.

_____

_____

Paper 2 — 45 minutes — 50 marks

1 The gym gives its members a leaflet about healthy eating. **[1]**

(a) Which nutrient is needed for strong bones and teeth?

- ☐ A) iron
- ☐ B) calcium
- ☐ C) vitamin C
- ☐ D) fat

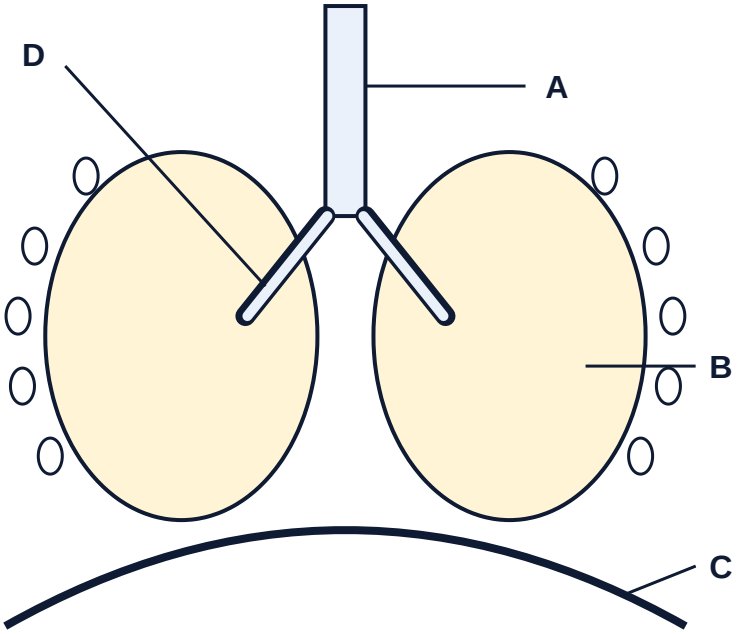
(b) A weightlifter wants to build more muscle. Which nutrient is needed for growth and repair of the body? **[1]**

(c) Animals can store energy in their bodies. Which **two** nutrients are used as energy stores? **[1]**

- ☐ A) vitamins and minerals
- ☐ B) protein and water
- ☐ C) calcium and iron
- ☐ D) carbohydrates and fats

(d) The leaflet also warns gym members not to smoke. **[2]**
Describe **two** ways smoking can affect a person's health.

2 The diagram shows the human breathing system. A gym trainer uses it to explain what happens when we breathe. **[1]**



(a) Which letter shows the **diaphragm**?

(b) Which describes what happens when we breathe **in**? **[1]**

- ☐ A) the diaphragm relaxes and moves up; the volume of the chest decreases
- ☐ B) the diaphragm contracts and moves down; the volume of the chest increases
- ☐ C) the intercostal muscles relax and the ribs move down and in
- ☐ D) the diaphragm contracts and moves up; the volume of the chest decreases

(c) Describe how oxygen moves from the air in the alveoli into the blood. Use the word **diffusion** in your answer. **[2]**

(d) A gym member says: "Breathing and respiration are the same thing." **[2]**
Explain why he is wrong.

3 On another island, farmers sprayed a pesticide on their fields. Rain washed some pesticide into the sea. **[1]**

Scientists measured the concentration of the pesticide in organisms in a sea food chain.

Organism	Pesticide concentration in units
algae	0.04
small shrimps	0.4
small fish	2
seabirds	20

(a) Which organism has the highest concentration of pesticide?

(b) How many times greater is the concentration in the seabirds than in the small fish? **[1]**

Working

Answer ____ times

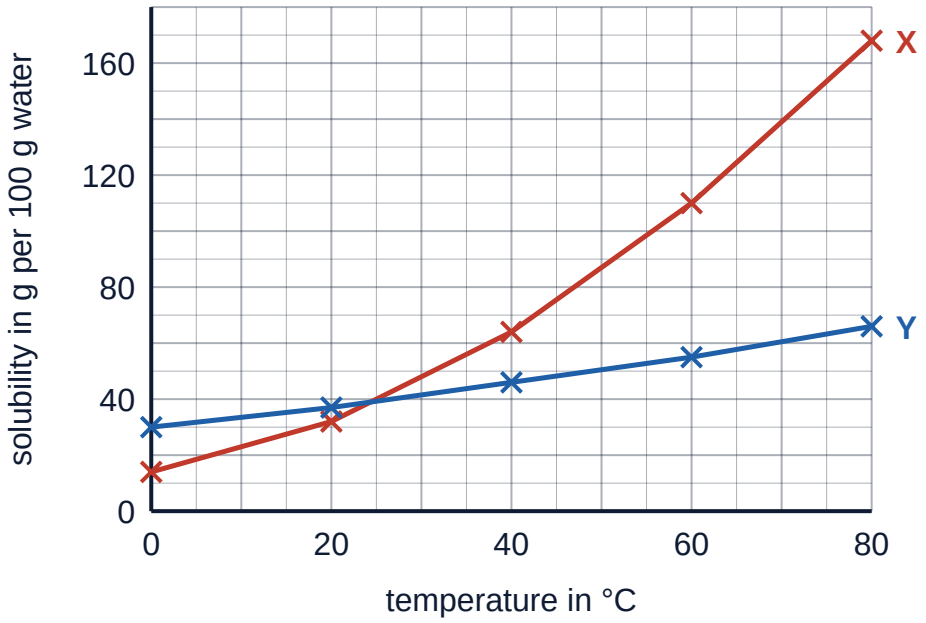
(c) Explain why the concentration of pesticide increases along the food chain. **[2]**

4 Scientists at the fertiliser factory investigate how the solubility of two salts, **X** and **Y**, changes with temperature. **[1]**

At each temperature they find the largest mass of salt that will dissolve in 100 g of water.

Temperature in °C	0	20	40	60	80
Salt X: solubility in g per 100 g water	14	32	64	110	168
Salt Y: solubility in g per 100 g water	30	37	46	55	66

The graph shows the results. The points are joined with straight lines.



(a) Name the **dependent** variable in this investigation.

(b) Describe how the solubility of salt X changes as the temperature increases. **[1]**

(c) Use the graph. At what temperature do salts X and Y have the **same** solubility? **[1]**

Working

Answer °C

(d) A worker makes a saturated solution of salt X in 100 g of water at 60 °C. She then cools it to 20 °C. **[2]**

Calculate the mass of crystals of X that form. Show your working.

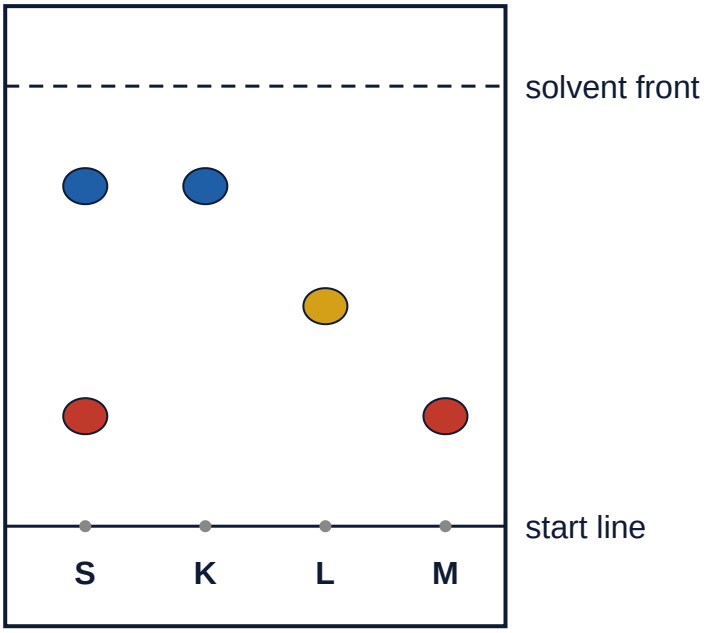
Working

Answer ____ g

(e) The scientists heat the water with a hot plate. Describe **one** risk and how to control it. **[1]**

5 The factory adds coloured dyes to its fertilisers so that farmers can tell them apart. **[1]**

A scientist checks the dye in a fertiliser sample, **S**, using paper chromatography. She compares it with three known dyes, **K**, **L** and **M**.



(a) Which dyes are in sample S?

- ☐ A) K only
- ☐ B) K and L
- ☐ C) K and M
- ☐ D) L and M

(b) Explain why the start line is drawn in pencil, not in ink. **[1]**

(c) What does the chromatogram show about sample S? **[1]**

- ☐ A) S is a pure substance
- ☐ B) S is a mixture of two dyes
- ☐ C) S contains dye L
- ☐ D) S does not dissolve in the solvent

6 Ammonium nitrate is a fertiliser. A student dissolves some ammonium nitrate in water. **[1]**
The temperature of the water is 20 °C at the start and 12 °C at the end.

(a) By how many degrees does the temperature fall?

Working

Answer ____ °C

(b) Is dissolving ammonium nitrate **exothermic** or **endothermic**? **[1]**

(c) The factory engineer looks at how four metals react with cold water. **[1]**

Metal	Reaction with cold water
calcium	reacts steadily, giving bubbles of gas
copper	no reaction
magnesium	a few bubbles form very slowly
sodium	reacts very quickly, fizzes and moves about on the surface

Which list shows the metals in order of reactivity, **most** reactive first?

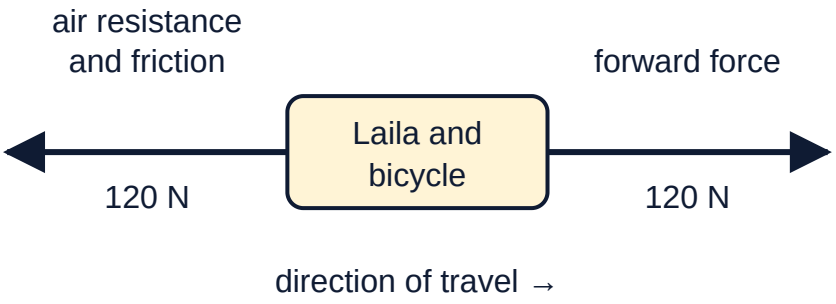
- ☐ A) sodium, calcium, magnesium, copper
- ☐ B) calcium, sodium, copper, magnesium
- ☐ C) copper, magnesium, calcium, sodium
- ☐ D) sodium, magnesium, calcium, copper

(d) Gold is used to coat electrical contacts in the factory's control room because it is **inert**. What does inert mean? **[1]**

- ☐ A) it reacts quickly with water
- ☐ B) it is magnetic
- ☐ C) it is generally unreactive
- ☐ D) it dissolves in water

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7 Laila cycles to school. The diagram shows the horizontal forces on Laila and her bicycle on a flat road. **[1]**



(a) Laila is already moving. Describe her motion while these forces act.

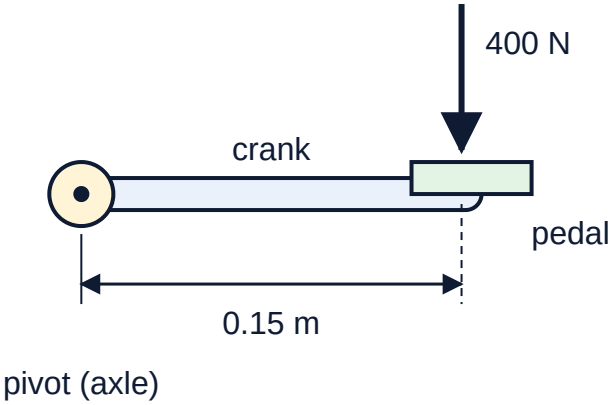
- ☐ A) she speeds up
- ☐ B) she slows down
- ☐ C) she stops at once
- ☐ D) she moves at a steady speed

(b) At the bottom of a hill Laila pedals harder. The forward force becomes 150 N. The backward force stays at 120 N. **[1]**
Calculate the resultant force.

Working

Answer _____ N

[2]



(c) The diagram shows Laila's pedal crank. She pushes down on the pedal with a force of 400 N, 0.15 m from the pivot.
Calculate the turning moment. Show your working.

Working

Answer _____ N m

(d) Laila and her bicycle have a total weight of 800 N. The two tyres touch the road over a total area of 16 cm^2. **[2]**
Calculate the pressure on the road. Show your working.

Working

Answer _____ N/cm^2

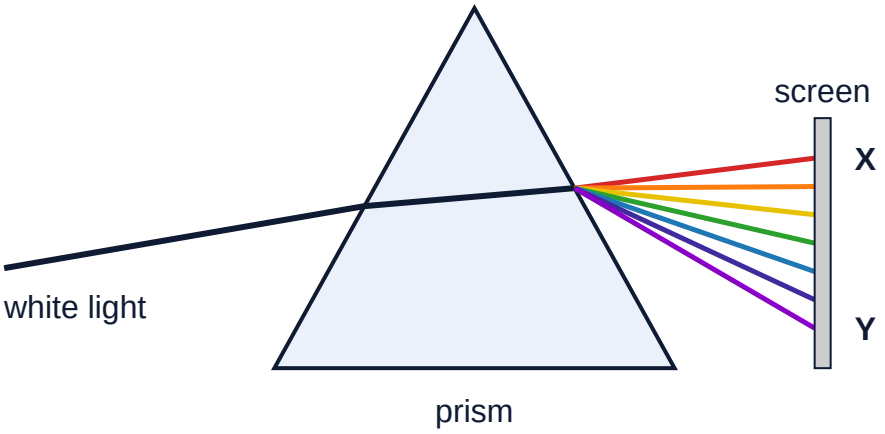
(e) Laila pumps more air into her tyre. Use ideas about particles to explain why the pressure of the air inside the tyre increases. **[2]**

8 During a show at a theatre, an actor sprays perfume on the stage. A lighting designer tests lights with a glass prism. **[1]**

(a) After a few minutes, people in the front row can smell the perfume. Name the process by which the perfume particles spread through the air.

(b) The designer shines a ray of light from air into a glass block. The ray bends as it enters the glass. **[1]**
Explain why light bends (refracts) at the boundary.

[1]



(c) The diagram shows white light passing through the prism and making a spectrum on a screen. Name the colour seen at **Y**.

(d) A yellow hat reflects red and green light. On stage, only **blue** light shines on the hat. What colour does the hat look? **[1]**

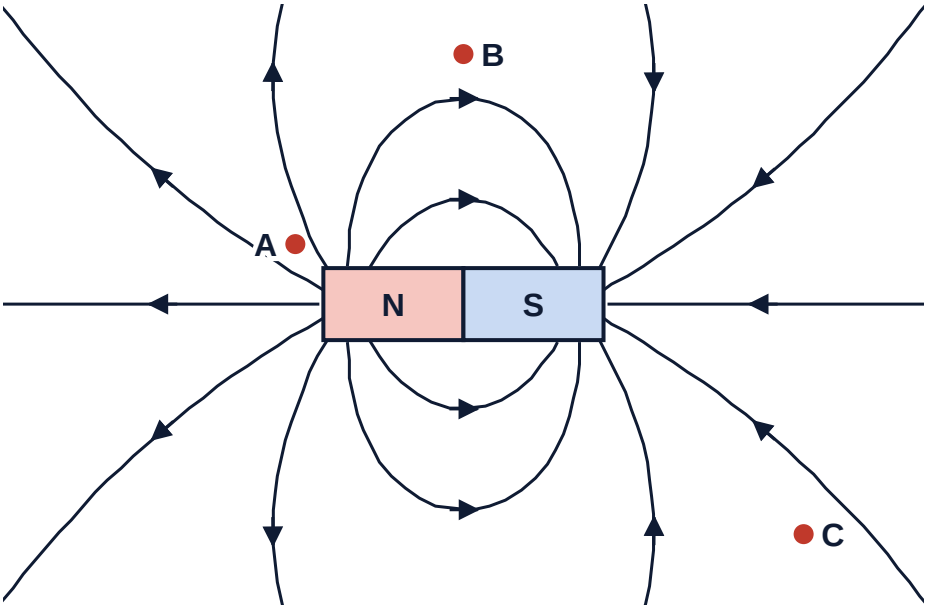
- ☐ A) yellow
- ☐ B) blue
- ☐ C) black
- ☐ D) green

9 A spacecraft is travelling to study an asteroid. **[1]**

(a) Which statement about asteroids is correct?

- ☐ A) asteroids are rocks that are smaller than planets
- ☐ B) asteroids are small stars
- ☐ C) asteroids are larger than planets
- ☐ D) asteroids are made only of gas

[1]



(b) The spacecraft carries a bar magnet. The diagram shows the magnetic field around it.
At which point, **A**, **B** or **C**, is the magnetic field strongest?

(c) Engineers test an electromagnet for the spacecraft. They change the number of turns on the coil and count how many paper clips it can hold. **[1]**

Number of turns	10	20	30	40
Number of paper clips	4	8	12	16

Predict how many paper clips the electromagnet will hold with 50 turns.

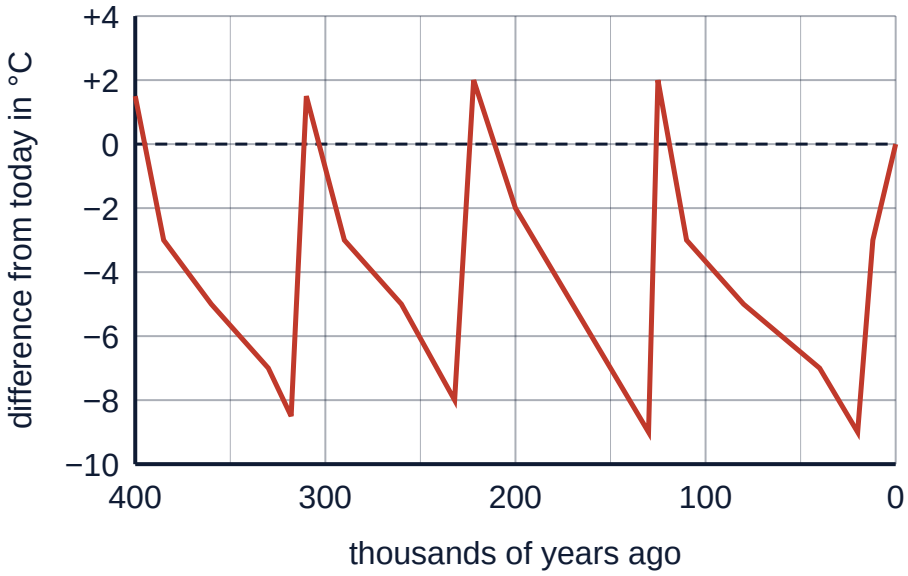
Working

Answer ☐ paper clips

(d) Give **one** other way the engineers could make the electromagnet stronger, without changing the number of turns. **[1]**

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10 In a climate data project, students study a graph made from ice-core data. It shows how much colder or warmer than today the Earth's average temperature was over the last 400 000 years. (The graph is simplified.) **[1]**



(a) Use the graph. About how many years pass from one warm period to the next? Give your answer in years.

Working

Answer ____ years

(b) What name is given to the long, very cold periods shown on the graph (about 8 °C colder than today)? **[1]**

(c) Scientists say the Earth's climate is warming now because the atmosphere is changing. Explain how burning fossil fuels can change the climate. **[2]**

(d) Which of these energy resources is **non-renewable**? **[1]**

- ☐ A) wind
- ☐ B) tidal
- ☐ C) solar
- ☐ D) coal

(e) A student finds a website about climate change that is paid for by an oil company. Explain why she should be careful when using information from it. **[1]**